

Diffusion Model

An introduction

Yukinari Hisaki

Contents

1. Motivation and background
2. Score-Based Generative Modeling with SDEs

- The purpose of this talk is to understand the Diffusion Model, which is part of the Diffusion Planner algorithm.
- Diffusion Model is more difficult than previous talk about Transformer. Diffusion Model is based on the difficult mathematical theory.
- However, the outcome of this theory is a simple algorithm.
- A rough understanding of the theory is sufficient; you don't need to worry about the details.

Motivation and background

The central goal is to estimate an unknown data distribution $q(x)$ from samples $x \sim q(x)$.

- **The Standard Approach (e.g., VAEs):** Introduce a latent variable $z \sim \mathcal{N}(0, I)$ and map it to data via a **single** probabilistic decoding step $p_{\theta(x|z)}$.

$$z \xrightarrow{p_{\theta(x|z)}} x$$

- **The Limitation:** Modeling the complex data distribution $q(x)$ via a “**single shot**” transformation $p_{\theta(x|z)}$ is challenging. The gap between the simple prior and the complex data manifold is too large.
- **The Solution (Diffusion Models):** Decompose the generation into **multiple gradual transformations**. We refine the sample step-by-step. $x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$

$$x_T \sim \mathcal{N}(\cdot \mid 0, I), x_0 \sim q(x_0)$$

The idea of “gradual transformation” can be mathematically formulated in three main ways:

1. Denoising Diffusion Probabilistic Models (DDPM) [1]

- Formulated as discrete Markov chains.
- Optimized via Variational Lower Bound (ELBO).
- Key idea: Learn to reverse the noise addition step-by-step.

2. Score-based Generative Models [2]

- Formulated via Stochastic Differential Equations (SDEs).
- Key idea: Learn the **score function** $\nabla_x \log p_{t(x)}$ (direction to higher density).

3. Flow Matching

- A more general framework based on Ordinary Differential Equations (ODEs).
- Key idea: Directly regress a **vector field** $v_t(x)$ to transport probability paths.

Focus of this talk: We will focus on **2. Score Matching** and **3. Flow Matching**, as they offer a unified continuous-time perspective.

Score-Based Generative Modeling with SDEs

Before introducing diffusion, let's define the “Score”.

- **Definition: The Stein Score Function** For a probability density $q(x)$, the score is defined as the **gradient of the log-density** with respect to data x :

$$s(x) \stackrel{\text{def}}{=} \nabla_x \log q(x)$$

- **Intuition: A Vector Field**

- ▶ The score is a vector field pointing in the direction of **higher data density**.
- ▶ At the modes (peaks) of the distribution, the score is zero ($\nabla \log q(x) = 0$).

- **The Goal (Score Matching)** We want to train a neural network $s_{\theta(x)}$ to approximate this true score:

$$\mathcal{L}_{\text{SM}}(\theta) = \mathbb{E}_{x \sim q} [\| s_{\theta}(x) - \nabla_x \log q(x) \|_2^2]$$

Problem: The true score $\nabla_x \log q(x)$ is intractable.

The Tractable Objective: We minimize the expected distance between the model and the **conditional score** of an auxiliary distribution $q(\tilde{x}|x)$:

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{q(x)q(\tilde{x}|x)} [\| s_{\theta}(\tilde{x}) - \nabla_{\tilde{x}} \log q(\tilde{x}|x) \|_2^2]$$

What is being optimized? Minimizing $\mathcal{L}_{\text{DSM}}$ is equivalent to minimizing the **Fisher Divergence** between the model and the **smoothed marginal** $q(\tilde{x}) = \int q(\tilde{x}|x)q(x)dx$.

$$\Rightarrow s_{\theta}(\tilde{x}) \approx \nabla_{\tilde{x}} \log q(\tilde{x})$$

Choice of Conditional Distribution: Typically, we choose a **Gaussian with small variance** for $q(\tilde{x}|x)$ (e.g., $\mathcal{N}(\tilde{x}; x, \sigma^2 I)$). This allows $q(\tilde{x})$ to approximate the data distribution $q(x)$ closely while keeping the objective tractable.

Once we have the score function $s_{\theta(x)} \approx \nabla_x \log q(x)$, how do we generate samples?

- **Langevin Dynamics (Noisy Gradient Ascent):** We initialize x_0 arbitrarily and iteratively update it using the score:

$$x_{t+1} \leftarrow x_t + \underbrace{\varepsilon_t s_{\theta}(x_t)}_{\text{Gradient Ascent}} + \underbrace{\sqrt{2\varepsilon_t} z_t}_{\text{Random Noise}}$$

where $z_t \sim \mathcal{N}(0, I)$ and ε is the step size.

- **Mechanism:**
 1. **Gradient Term:** Pushes x towards high-density regions (modes).
 2. **Noise Term:** Prevents collapsing to a single mode, allowing exploration of the full distribution.

$\Rightarrow$ As $t \rightarrow \infty$ and $\varepsilon_t \rightarrow 0$, x_t converges to a sample from $q(x)$.

Theory is sound, but training on raw data $x \sim q(x)$ often fails in practice.

1. The Manifold Hypothesis (Blindness)

- High-dimensional data concentrates on a low-dimensional manifold.
- Outside this manifold, $q(x) \approx 0$.
- The score $\nabla_x \log q(x)$ is **undefined** or numerically unstable in the ambient space.

2. Inaccurate Estimation in Low-Density Regions

- The loss $\mathbb{E}_{x \sim q}$ only minimizes error **where data exists**.
- In low-density regions (most of the space), the model $s_{\theta(x)}$ is untrained.
- **Result:** When sampling starts from random noise (low density), the gradients are inaccurate, leading to garbage generation.

3. Slow Mixing of Langevin Dynamics

- If the data distribution has disjoint modes (isolated “islands” of data), the score in between is near zero.
- The sampler gets stuck in one mode and cannot cross the gap to others.

References

- [1] “Denoising Diffusion Probabilistic Models.” [Online]. Available: <https://arxiv.org/abs/2006.11239>
- [2] “Score-Based Generative Modeling through Stochastic Differential Equations.” [Online]. Available: <https://arxiv.org/abs/2011.13456>